



STUDENT HANDOUT

The Maker's Space – Local Flight Radar
Waveshare ESP32-S3-Touch-LCD-1.46C · Arduino IDE

What You're Building

A self-contained, battery-capable flight radar that pulls live ADS-B data from adsb.fi over WiFi and displays nearby aircraft as glowing blips on a 1.46" round display. Doubles as a clock, has multiple themes, a web settings portal, OTA firmware updates, optional flight logging, and a demo mode for showing it off without WiFi.

Required Hardware

This class uses the **Waveshare ESP32-S3-Touch-LCD-1.46C** board.

Why this specific board? It integrates the display, touch, IMU, RTC, and audio on one compact board - perfect for a portable flight radar. While the code can support other hardware (see [board_config.h](#)), this class focuses on this specific integrated solution.

The Board at a Glance

Component	Chip	Purpose
MCU	ESP32-S3 (240 MHz, dual-core, WiFi+BT)	Runs everything
Display	SPD2010, 412×412 round, QSPI	Radar + UI
Touch	SPD2010 (same chip), I ² C	Gestures
IMU	QMI8658 6-axis	Auto-rotate + shake-to-wake
RTC	PCF85063 with coin-cell backup	Keeps time without WiFi
Audio	PCM5101 DAC + speaker	Alert chirps
Storage	16 MB Flash + 8 MB PSRAM + microSD slot	Code, framebuffer, logs
Other	TCA9554 I/O expander, microphone, battery charger	—

For detailed specifications, pinout diagrams, and schematic:

Waveshare ESP32-S3-Touch-LCD-1.46 Documentation (<https://docs.waveshare.com/ESP32-S3-Touch-LCD-1.46>)

Flashing Your Board

Follow the [Build Guide](#) to:

1. Download the code from GitHub
2. Install Arduino IDE, board support, and libraries
3. Configure board settings and upload
4. Verify the display shows "SETUP"

Return here when your board shows the SETUP screen and continue with First-Time Setup below.

First-Time Setup

1. **Power on** your already-flashed device. The screen shows **SETUP** and a unique WiFi name like **FLightRadar-A1B2**.
2. **Join that WiFi** from your phone. The captive portal opens automatically — if not, browse to **192.168.4.1**.
3. **Pick your home WiFi**, enter your password, and fill in:
 - Home latitude / longitude (default: Fort Wayne, IN)
 - Timezone offset from UTC (e.g., **-5** for EST)
4. **Save**. The device reconnects to your WiFi and the radar appears.
5. After connection, browse to **http://flightradar.local/** on the same network for full settings.

Touch Gestures

Gesture	What Happens
Tap a blip	Open the flight detail panel
Tap in flight detail	Return to radar
Tap a settings row	Cycle that value (theme, range, units, etc.)
Long-press (>500 ms)	Open the settings overlay
Swipe left (radar)	Go to Stats
Swipe down (radar)	Go to Clock
Swipe right (most screens)	Back to radar
Drag finger around the outer ring	Adjust brightness (clockwise = brighter)
Shake the device	Wake the screen from dim mode

On-Screen Symbols

- | | |
|------------------------|----------------------------------|
| ▲ Heavy jet (airliner) | ◆ Medium jet (regional/business) |
| ✚ Helicopter | ⊙ Military |
| ○ Light aircraft (GA) | △ Glider (outline only) |

A line trailing from each blip shows direction and speed.
 A short tail of dim blips behind the sweep = phosphor afterglow.

Colors Tell You Altitude

Color brightness	Altitude
Dim	Below 5,000 ft
Normal	5,000–18,000 ft
Bright	18,000–35,000 ft
Near-white	Above 35,000 ft

Emergency squawk overrides:

-  **7500** — Hijack
-  **7600** — Radio failure
-  **7700** — General emergency

If the radar plays three urgent beeps and a blip turns red, you're watching aviation history happen.

Web Portal (<http://flightradar.local/>)

Everything from the on-screen settings plus:

- Precise lat/lon entry
- Min altitude filter (filters out ground vehicles)
- Idle-dim timeout
- OTA password
- Reboot / forget-WiFi buttons

Settings Reference

Setting	Options	What it does
Theme	Green Phosphor / Amber CRT	Color palette
Range	5 / 10 / 25 / 50 (units)	Outer ring radius
Units	Native (nm/kt) / Miles (mi/mpH) / Metric (km/kmh)	Display units
Time	12h / 24h	Clock format
Audio	ON / OFF	Alert chirps
Demo	ON / OFF	Fake aircraft (no WiFi needed)

Setting	Options	What it does
Bright	10–255	Backlight level (also via edge drag)
Log	ON / OFF	Append spotted flights to <code>/flights.csv</code> on internal flash

Troubleshooting

Symptom	Try
Black screen, backlight on	Power-cycle (unplug 5 s). Confirm Tools → PSRAM = Enabled if reflashing.
Can't find AP / portal	Hold the BOOT button 5 s after boot — this clears saved WiFi.
No flights showing	Confirm WiFi is connected (top-right "WiFi" indicator green). Try enabling Demo mode to verify rendering.
Touch unresponsive	Restart. The SPD2010 needs a few seconds after boot to leave BIOS mode.
Time is wrong	Set timezone offset in the web portal. NTP runs once on connect.
Want a clean start	In web portal → "Forget WiFi" button.

Where to Go Next

- **Modify the code.** Try changing a theme color, adding a new range preset, or tweaking the sweep speed.
- **Add aircraft types.** Edit `aircraft_types.h` and add entries — they're alphabetically sorted (binary searched).
- **Add a third theme.** Copy the Amber CRT block in `themes.h`, change the colors, and bump `THEME_COUNT`.
- **Flash via WiFi.** With OTA enabled, Arduino IDE → Tools → Port shows `flightradar at <ip>` as a network port. No USB needed.
- **Tweak the layout.** Edit `ui_layout.h` to adjust button positions, gesture sensitivity, or the settings menu spacing.
- **Adjust the gesture feel.** Edit `ui_layout.h` and change `SWIPE_THRESHOLD` or `LONG_PRESS_DURATION_MS` to make touches more/less sensitive.
- **Move UI elements.** All screen positions live in `ui_layout.h` - try moving the WiFi icon or battery indicator.

Quick Code Tour

```
FlightRadar/
├─ FlightRadar.ino ← Main sketch (3 parts: init, render, input/network)
├─ board_config.h ← Every hardware pin lives here
├─ font_config.h ← Set font sizes
```

- |— ui_layout.h ← Screen positions and gesture thresholds
- |— themes.h ← Color palettes
- |— aircraft_types.h ← ICAO code → human name lookup

Class project page (with code, schematics, and updates):

<https://github.com/computethis/TheMakersSpace-Flight-Radar>

Built with: Arduino_GFX · ArduinoJson · WiFiManager
· SensorLib · adsb.fi API · tar1090-db (concept credit)
